Federico Tesone

Principal Engineer · Search, Retrieval & Agentic AI Systems

New York, NY, USA | fedetesone.github.io | linkedin.com/in/ftesone

Summary

Principal engineer with 14+ years building search, retrieval, and agentic AI systems. Leads Daydream's 9-person Search Core team, which took conversational search from lexical retrieval to hybrid retrieval with LLM orchestration, preference memory, and evaluation-gated releases. Previously set search strategy at Grubhub across 3 teams and 20 engineers on a platform serving 120K–150K requests per minute, delivering +2.37% conversion and $134K in annual Elasticsearch savings.

Experience

Principal Engineer, Daydream – New York, NY, USA July 2025 – present

Lead a 9-person Search Core team owning retrieval, ranking, and query understanding for an AI-native fashion search and conversational engine.

- Architected the search engine from the ground up — lexical to hybrid text, visual, and semantic retrieval across 5M+ product options — then extended it to image search, visual discovery, multi-turn preference memory, and embedding pipelines for personalization.
- Re-architected query understanding from a single blocking LLM call into a concurrent DAG with confidence routing and multi-intent splitting — cut P95 latency from 7s to 2.5s (−64%) despite substantially more signal extraction per query.
- Designed multi-pass re-ranking with agentic fallback loops, adding a planner/judge stage that re-ranks retrieved products with structured decision models (TypeSafe Jev) — raised the LLM-judged quality score from 0.66 to 0.75 and merchant click-out rate by 12%.
- Built the evaluation and observability backbone: offline/online evaluation gating ranking and prompt changes, plus end-to-end tracing of LLM decisions, retrieval stages, ranking signals, latency, and final results.

Staff Engineer, Grubhub – New York, NY, USA Feb 2023 – July 2025

Technical lead of an 8-engineer search team; set technical strategy and architecture across Search, Recommender Systems, and Embeddings (3 teams, 20 engineers).

- Architected a modular search runtime separating context, candidate selection, ranking, experimentation, and hydration, enabling independent evolution of retrieval and ML ranking systems.
- Operated and evolved Grubhub's search and discovery platform serving approximately 120K–150K requests per minute across major consumer surfaces.
- Improved search conversion by 2.37% through multi-query retrieval and result blending.
- Reduced Elasticsearch CPU utilization by 34% — $134K in annual compute savings — by redesigning retrieval from multi-query execution to a single-query architecture, while improving ranking flexibility and production scalability.
- Designed candidate selection and ranking infrastructure integrating real-time ML model predictions, embeddings, business constraints, geographic filtering, and experimentation signals.
- Mentored senior engineers across 3 teams.

Senior Software Engineer, Grubhub – New York, NY, USA Dec 2019 – Feb 2023

Engineered high-availability backend microservices for the discovery runtime platform.

- Built a low-latency vector retrieval system for production runtime workloads using Annoy, with custom sharding of vector indexes to support scalable embedding-based lookup across large datasets.
- Developed low-latency data serving infrastructure for embedding-based ML models and user targeting.
- Partnered with machine-learning and data teams to productionize new ranking and recommendation capabilities.

Senior Developer, Endava – Rosario, Argentina July 2016 – Dec 2019
Developed backend systems and microservices using Java, Scala, Python, Cassandra, AWS, and CI/CD pipelines.

Auxiliary Professor, Universidad Abierta Interamericana – Rosario, Argentina Aug 2014 – Mar 2019
Taught Object-Oriented Software Design, GoF Design Patterns, Software Metrics, QA, and Automated Testing.

Earlier Experience

Team Lead & Senior Developer, Nación Servicios (2013–2016): Core banking loyalty platforms, transaction authorization systems, and EMV payment services.

BI Consultant, BPD Solutions (2012–2013): Enterprise ETL pipelines, data integration workflows, and data warehousing systems.

Education

Universidad Tecnológica Nacional, MSc in Software Engineering 2013 – Nov 2021
Thesis: Developing a Similarity Measure for Recommender Systems on Community Question Answering Sites: A Big Data Analysis Using a Clustering Ensemble Method. Written in Spanish: *Desarrollo de una medida de similaridad para Sistemas de Recomendación en sitios de Community Question Answering. Análisis desde un enfoque Big Data y usando un método de ensamble de clustering.*

Universidad Tecnológica Nacional, BSc in Information Systems Engineering 2007 – 2012
5-year degree with minor thesis.

Technical Expertise

AI & Search Systems: Hybrid Retrieval, Dense & Sparse Retrieval, Vector Search, Ranking & Re-ranking, LLM Query Understanding, Agentic Search, Multi-turn Preference Memory, Evaluation Systems, Embeddings, Personalization, Recommendation Systems

Backend & Data Infra: High-Throughput, Low-Latency Services, Real-Time Data Serving, Feature Infrastructure, Event-Driven Systems, Observability, Reliability Engineering

Languages & Tech: Java, Scala, Python, SQL, Elasticsearch, Cassandra, ScyllaDB, Redis, Google Cloud, AWS

Languages: English (Full Professional), Spanish (Native)

Honors and Awards

Cuadro de Honor - Honor Roll
Published in the list of highest grade point averages of the Province of Santa Fe, Argentina.